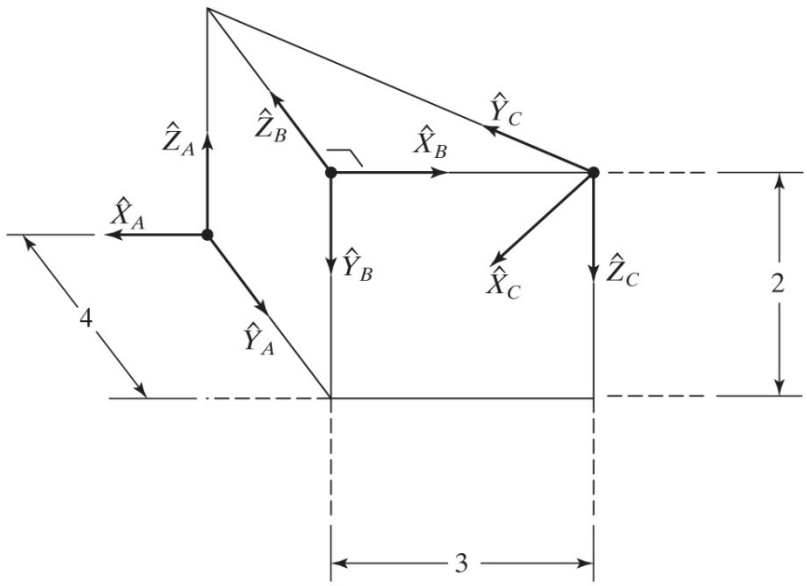
MSE 429/894 – Assignment 1

Problem 1: Consider the frames at the corners of a wedge as shown below. Find the value of ${}^{A}_{B}T$, ${}^{A}_{C}T$, ${}^{B}_{C}T$, and ${}^{C}_{A}T$.



Solution:

$$ {}^{A}_{B}T = \begin{bmatrix} {}^{A}_{B}R & & & {}^{A}P_{B_{ORG}} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

```
alpha = 180;
beta = 0;
gamma = 90;
T_AtoB =

   -1.0000   -0.0000         0         0
    0.0000   -0.0000   -1.0000    4.0000
    0.0000   -1.0000    0.0000    2.0000
         0         0         0    1.0000
```

```
alpha = 180 - atan(3/4;
beta = 0;
gamma = -90;
T_BtoC =

   -0.8000   -0.6000         0    3.0000
    0.0000   -0.0000    1.0000         0
   -0.6000    0.8000    0.0000         0
         0         0         0    1.0000
```

```
alpha = -atan(3/4);
beta = 0;
gamma = 180;
T_AtoC =

    0.8000    0.6000         0   -3.0000
    0.6000   -0.8000   -0.0000    4.0000
   -0.0000    0.0000   -1.0000    2.0000
         0         0         0    1.0000
```

```
alpha = -atan(3/4);
beta = 0;
gamma = 180;
T_CtoA =

    0.8000    0.6000         0         0
    0.6000   -0.8000   -0.0000    5.0000
   -0.0000    0.0000   -1.0000    2.0000
         0         0         0    1.0000
```

Problem 2: Determine

a) The rotation matrix, ${}^{A}_{B}R$, that describes frame $\{B\}$ relative to $\{A\}$ if

$$ {}^{A}\hat{X}_B = [1 \quad 0 \quad 0]^T $$
$$ {}^{A}\hat{Y}_B = [0 \quad 0 \quad -1]^T $$
$$ {}^{A}\hat{Z}_B = [0 \quad 1 \quad 0]^T $$

b) The $Z-Y-Z$ Euler angles required for such a rotation.

Solution:

a) Use

$$ {}^{A}_{B}R = \left[{}^{A}\hat{X}_B \; {}^{A}\hat{Y}_B \; {}^{A}\hat{Z}_B \right] = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} $$

to obtain

$$ {}^{A}_{B}R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} $$

b) Use

$$ \beta = \mathrm{A}\tan 2(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}), $$
$$ \alpha = \mathrm{A}\tan 2(r_{23}/s\beta, r_{13}/s\beta), $$
$$ \gamma = \mathrm{A}\tan 2(r_{32}/s\beta, -r_{31}/s\beta). $$

to get

$$ \alpha = 90 \text{ degrees} $$
$$ \beta = 90 \text{ degrees} $$
$$ \gamma = -90 \text{ degrees} $$

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Problem 3: A coordinate frame, $\{B\}$, is located at the base of a robot manipulator. $\hat{Z}_B$ points upward. Three cameras are used to view the manipulator. Coordinate frames $\{C\}$, $\{D\}$, and $\{E\}$ describe the camera positions and orientations. The origin of $\{C\}$ is on the $\hat{X}_B - \hat{Z}_B$ plane. The cameras are mounted on tripods 1.5 units tall placed at the vertices of an equilateral triangle having $\{B\}$ at the incenter. The robot frame is on the focal axis of each camera (the camera's $\hat{Z}$-axis), and the Euclidean distance from $\{B\}$ to any camera is 5 units, thus

$$^{C}P_{B_{ORG}} = {}^{D}P_{B_{ORG}} = {}^{E}P_{B_{ORG}} = [0 \quad 0 \quad 5]^T$$

Compute the camera transformation matrices: ${}^{B}_{C}T$, ${}^{B}_{D}T$, and ${}^{B}_{E}T$.

Solution:

Use

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \left[\begin{array}{c|c} {}^{A}_{B}R & {}^{A}P_{BORG} \\ \hline 0 \quad 0 \quad 0 & 1 \end{array}\right] \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix}$$

to obtain the transformation matrices. The rotation is X-Y-Z fixed angles, so use

$${}^{A}_{B}R_{XYZ}(\gamma,\ \beta,\ \alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

for that 3×3 submatrix, with angles

$$\gamma = 0 \text{ degrees}$$

$$\beta = -\sin^{-1}\left(\frac{\text{tripod_height}}{\text{distance_along_optical_axis}}\right) = -\sin^{-1}\left(\frac{1.5}{5}\right) = -107 \text{ degrees}$$

$$\alpha_C = 0 \text{ degrees}$$

$$\alpha_D = 120 \text{ degrees}$$

$$\alpha_E = 240 \text{ degrees}$$

The position vectors to the camera-frame origins are

$${}^{B}P_{CORG} = \begin{bmatrix} \text{horizontal_distance} \\ 0 \\ \text{tripod_height} \end{bmatrix} = \begin{bmatrix} 4.77 \\ 0 \\ 1.50 \end{bmatrix}$$

$${}^{B}P_{DORG} = \begin{bmatrix} \text{horizontal_distance} \times \cos\alpha_D \\ \text{horizontal_distance} \times \sin\alpha_D \\ \text{tripod_height} \end{bmatrix} = \begin{bmatrix} -2.39 \\ 4.13 \\ 1.5 \end{bmatrix}$$

$${}^{B}P_{EORG} = \begin{bmatrix} \text{horizontal_distance} \times \cos\alpha_E \\ \text{horizontal_distance} \times \sin\alpha_E \\ \text{tripod_height} \end{bmatrix} = \begin{bmatrix} -2.38 \\ -4.13 \\ 1.50 \end{bmatrix},$$

where $\text{horizontal_distance} = \sqrt{(\text{distance_along_optical_axis})^2 - (\text{tripod_height})^2}$.

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Combining the rotation and translation yields the transformation matrices via

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \left[\begin{array}{ccc|c} & {}^{A}_{B}R & & {}^{A}P_{BORG} \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix}$$

as

$${}^{B}_{C}T = \begin{bmatrix} -.300 & 0 & -.954 & 4.77 \\ 0 & 1.00 & 0 & 0 \\ .954 & 0 & -.300 & 1.50 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^{B}_{D}T = \begin{bmatrix} .150 & -.866 & .477 & -2.39 \\ -.260 & -.500 & -.826 & 4.13 \\ .954 & 0 & -.300 & 1.50 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^{B}_{E}T = \begin{bmatrix} .150 & .866 & .477 & -2.39 \\ .260 & -.500 & .826 & -4.13 \\ .954 & 0 & -.300 & 1.50 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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Problem 4: The position and velocity of an object are known to be ${}^{B}P_0 = [0 \quad 0.5 \quad 0]^T$ and ${}^{B}V_0 = [1.9 \quad 0.1 \quad -0.3]^T$ at time t_0, respectively. If the velocity is constant and

$$ {}^{A}_{B}T = \begin{bmatrix} 0.0722 & -0.963 & -0.259 & -5.00 \\ 0.954 & -0.00868 & 0.298 & -6.50 \\ -0.290 & -0.269 & 0.919 & 8.00 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

then what is ${}^{A}P$ after 5 units of time?

Solution:

Let

$$ {}^{B}P_1 = {}^{B}P_0 + 5\,{}^{B}V_0 = [9.5 \quad 1.00 \quad -1.50]^{\mathsf{T}} $$

The object's position in $\{A\}$ is

$$ {}^{A}P_1 = {}^{A}_{B}T\,{}^{B}P_1 = [-4.89 \quad 2.11 \quad 3.60]^{\mathsf{T}} $$

Problem 5: A vector ${}^{A}P$ is rotated about $\hat{Z}_A$ by θ degrees, and is subsequently rotated about $\hat{Y}_A$ by ϕ degrees. Give the rotation matrix that accomplishes theses rotations in the given order.

Solution:

$$ \begin{aligned} R &= \mathrm{rot}(\hat{Y}, \phi)\,\mathrm{rot}(\hat{Z}, \theta) \\ &= \begin{bmatrix} c\phi & 0 & s\phi \\ 0 & 1 & 0 \\ -s\phi & 0 & c\phi \end{bmatrix} \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c\phi\,c\theta & -c\phi\,s\theta & s\phi \\ s\theta & c\theta & 0 \\ -s\phi\,c\theta & s\phi\,s\theta & c\phi \end{bmatrix} \end{aligned} $$

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Problem 6: A velocity vector is given by ${}^{B}V = [30.0 \quad 40.0 \quad 50.0]^{T}$. Given

$$ {}^{A}_{B}T = \begin{bmatrix} 0.707 & 0 & -0.707 & 11.0 \\ -0.612 & 0.500 & -0.612 & -3.0 \\ 0.353 & 0.866 & 0.353 & -9.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

compute ${}^{A}V$.

Solution:

Velocity is a "free vector" and only will be affected by rotation, and not by translation:

$$ {}^{A}V = {}^{A}_{B}R\,{}^{B}V = \begin{bmatrix} .707 & 0 & -.707 \\ -.612 & .500 & -.612 \\ .353 & .866 & .353 \end{bmatrix} \begin{bmatrix} 30.0 \\ 40.0 \\ 50.0 \end{bmatrix} $$

$$ = [-14.1 \quad -29.0 \quad 62.9]^{\mathsf{T}} $$

Problem 7: Given

$$ {}^{A}_{B}T = \begin{bmatrix} 0.25 & 0.43 & 0.86 & 5.0 \\ 0.87 & -0.50 & 0.00 & -4.0 \\ 0.43 & 0.75 & -0.5 & 3.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

what is ${}^{B}P_{A_{ORG}}$?

Solution:

Using

$$ {}^{B}_{A}R = {}^{A}_{B}R^{T}. $$

$$ {}^{B}({}^{A}P_{BORG}) = {}^{B}_{A}R\ {}^{A}P_{BORG} + {}^{B}P_{AORG}. $$

$$ {}^{B}P_{AORG} = -{}^{B}_{A}R\ {}^{A}P_{BORG} = -{}^{A}_{B}R^{T}\,{}^{A}P_{BORG}. $$

$$ {}^{B}_{A}T = \left[\begin{array}{ccc|c} & {}^{A}_{B}R^{T} & & -{}^{A}_{B}R^{T}\,{}^{A}P_{BORG} \\ \hline 0 & 0 & 0 & 1 \end{array}\right]. $$

$$ {}^{B}_{A}T = {}^{A}_{B}T^{-1} $$

we get that

$$ {}^{B}P_{AORG} = -{}^{A}_{B}R^{\mathsf{T}}\,{}^{A}P_{AORG} = -\begin{bmatrix} .25 & .87 & .43 \\ .43 & -.50 & .75 \\ .86 & .00 & -.50 \end{bmatrix} \begin{bmatrix} 5.0 \\ -4.0 \\ 3.0 \end{bmatrix} = \begin{bmatrix} .94 \\ -6.4 \\ -2.8 \end{bmatrix} $$

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